

Paper Manager pack — I-JEPA (Nov 25 · Accompanying Models)

Paper: Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA)
arXiv: <https://arxiv.org/abs/2301.08243> · [code](#) · **CVPR 2023** (arXiv v3, 13 Apr 2023)
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PDF: </workspace/papers/ijepa-2301.08243.pdf> (= </workspace/ijepa-quiet/paper.pdf>) · **17 pp** · arXiv **2301.08243v3**
Session: Nov 25 · due 9:30 AM · Type: **Accompanying Models (World Models, Reasoning, Planning)**
Role weights: **Archaeologist=1 (Kanta Saito) LONGEST PRIMARY**; **Stakeholder=2 (Lukas Moroz)**; Reviewer / Empiricist / Visionary seats empty — **still all five produced**; Archaeologist heaviest
Kanta role: Produce **Archaeologist-richest** five-role pack; deepen CLEAR/TENTATIVE JEPA lineage → V-JEPA → V-JEPA 2-AC; world-model / planning deep dive; SSL taxonomy; Empiricist course-smoke run pack
Themes: predict in embedding space (not pixels) · multi-block masking · EMA target encoder · no hand-crafted view augs · JEPA vs MAE vs JEA · efficiency vs MAE · Clevr Dist · lineage CLEAR to V-JEPA → V-JEPA 2-AC robotics (do NOT claim robot results from this paper)
Sources: Summarizer (</workspace/papers/ijepa-2301.08243-summary.md>) + Senior Research (</workspace/ijepa-quiet/DELIVERABLE.md>) + Ablation Bot (</workspace/papers/ijepa-2301.08243-ablations.md>) — **AUTHORITATIVE**) + [paper_notes.md](#) / [env_outline/](#)
Sibling links: </workspace/siglip-quiet/> · </workspace/paligemma-quiet/> · </workspace/openvla-quiet/> · </workspace/pi05-quiet/> · </workspace/openpi-comet-smoke/> · </workspace/diffusion-policy-quiet/> · </workspace/droid-quiet/> · </workspace/data-scaling-laws-quiet/> · </workspace/oxe-quiet/>

1. One-liner

Predict missing parts in embedding space, not pixels — from one **context** block, predict **EMA target-encoder** embeddings of several **multi-block** targets (no SimCLR-style view augs); multi-block is the critical inductive bias (**54.2** vs ~**15–20** on IN-1% lin); emb targets $\gg$ pixels (**66.9** vs **40.7**); image birth certificate of Meta’s JEPA line → **V-JEPA** → **V-JEPA 2-AC** (robot planning is **CLEAR subsequent**, not this PDF).

Contrast poles (consistency triangle):

Pole	What they do	Primary claim / use
DINOv2 / SigLIP	Discriminative JEA / contrastive–sigmoid towers	Dense / language-aligned features → VLA vision towers (π_0 , OpenVLA, ...)
MAE	Generative MIM — reconstruct pixels	Strong FT; weaker frozen linear; texture-rich
I-JEPA → V-JEPA → V-JEPA 2-AC	Predictive embedding / latent world model	Semantic SSL without view augs → video → action-conditioned planning

Do NOT invent / claim: robot SR, BEHAVIOR q-score, Franka pick-place, or V-JEPA 2-AC planning numbers **from this 2023 image paper**. Cite **V-JEPA 2-AC** for robotics. Do not claim π_0 / OpenVLA / GR00T cite or use I-JEPA (they don’t — SigLIP/DINOv2 won the tower market).

2. Summary (exec overview + key numbers)

Assran et al. (CVPR 2023 / Meta FAIR) instantiate LeCun’s JEPA on images: learn highly **semantic** representations without hand-crafted multi-view augmentations by predicting **target-block embeddings** from a single **context** block. Architecture = **context ViT** (sparse, MAE-style visible patches) + **EMA target ViT** (full image; mask applied to outputs) + **narrow predictor ViT** (emb **384**; positional mask tokens). Loss = patch-wise **L2 in embedding space**.

Why Accompanying Models session: Supplies the **foundational** non-generative SSL brick whose **CLEAR** Meta follow-ons are **V-JEPA** (video feature prediction) and **V-JEPA 2 / 2-AC** (action-conditioned world model + zero-shot

Franka planning). This paper itself has **zero** robotics / planning / action experiments — value is **idea + lineage + mask/target recipe**.

Axis	Number / claim
Venue / code	CVPR 2023; facebookresearch/ijepa CLEAR
IN-1K linear (T1)	ViT-H/14 79.3% (300 ep); ViT-H/16@448 81.1% (300 ep)
IN-1% (T2)	H/14 73.3%; H/16@448 77.3%
Transfer H/14 (T3)	CIFAR100 87.5 · Places205 58.4 · iNat18 47.6
Local H/14 (T4)	Clevr/Count 86.7 · Clevr/Dist 72.4 (<< DINO Dist 53.4)
Multi-block vs alts (T6)	54.2 vs raster 15.5 / block 20.2 / random 17.6
Emb vs pixels (T7)	66.9 vs 40.7 (ViT-L/16; 500 vs 800 ep)
Output vs input mask (T11)	67.3 vs 56.1
Predictor depth 12 vs 6 (T12)	66.9 vs 64.0
Full FT (T15)	I-JEPA H/16@448 87.1% (300 ep) vs MAE H/14@448 87.8% (1600 ep)
Compute	ViT-H/14 <1200 GPU-h; 16×A100 <72 h; ~10× cheaper than MAE H/14; ~2.5× faster than iBOT S/16
Per-iter vs MAE	~7% slower/iter; ~5× fewer epochs → net win
Default mask	M=4 targets scale (0.15, 0.2); context (0.85, 1.0); avg context ratio ~0.25
Collapse stop	EMA mom 0.996 → 1.0; asymmetric predictor; no InfoNCE

Must-cite tattoo: multi-block 54.2 vs ~15–20 · emb vs pixels 66.9 vs 40.7 · IN1K H/14 79.3 / H/16@448 81.1 · 1% 73.3 / 77.3 · Clevr Dist 72.4 · compute <1200 GPU-h / 16×A100 <72 h / ~10× vs MAE · V-JEPA 2-AC as CLEAR subsequent for robots.

Do NOT claim: robot results from this PDF; course smoke = 79% H/14; π_0 /OpenVLA use I-JEPA; “no augs” means no crop-like block sampling (mask sampling is the inductive bias).

3. Keywords

I-JEPA; Joint-Embedding Predictive Architecture; self-supervised learning; Vision Transformer; multi-block masking; representation-space prediction; EMA target encoder; masked image modeling; non-generative SSL; world models; predictive embeddings; energy-based models; LeCun AMI; V-JEPA; V-JEPA 2-AC; MAE foil; DINOv2/SigLIP foil; Clevr Dist; CVPR 2023; BEHAVIOR backbone ablation; Accompanying Models

4. Deep dive — World-model / planning / predictive embeddings (PRIMARY SESSION CONTRIBUTION)

Callout: Scientific product of *this* PDF = **static within-image latent prediction** (parts from context). Session type = Accompanying Models because the **JEPA program** is Meta’s explicit path to **latent world models** and later **action-conditioned planning**. Treat robotics claims as **lineage**, not in-paper results.

What is “world-model-ish” here — CLEAR

Piece	In-paper role	World-model reading
Predictor (g_{ϕ})	Narrow ViT; context tokens + positional mask tokens → predicted target embeddings	Conditional latent predictor ($P(\text{latent}_{target} \mid \text{latent}_{context}, z_{\text{pos}})$) — spatial “future/parts,” not temporal
Loss in emb space	L2 on EMA target patches	Discard unpredictable pixel noise; keep predictable structure (pose / parts — Fig. 6 RCDM)
Multi-block targets	Several large-ish semantic regions	Forces prediction of object-level content, not texture fill
No view-aug JEA	Single view	Prior = mask geometry , not invariance to color/crop stacks
LeCun AMI [48]	Cited parent	JEPA = predictive embeddings + EBM agenda toward AMI

What is not in this paper

- No temporal dynamics, video, proprioception, or actions.
- No CEM / MPC / planning loop.
- No robot, BEHAVIOR, DROID, or VLA experiments.
- Predictor visualizations (Sec 8 / Fig. 6) are **qualitative** RCDM decodes — suggestive, not quantitative planning proof.

CLEAR subsequent robotics path (cite these, not Assran’23 numbers)

Work	What it adds	Label
V-JEPA (Bardes et al. 2024, arXiv 2404.08471); <code>facebookresearch/jepa</code>	Same JEPA recipe on video ; feature prediction only; frozen video+image probes (e.g. ViT-H/16: K400 81.9 , SSv2 72.2 , IN1K 77.9)	CLEAR
V-JEPA 2 (Assran et al. 2025, arXiv 2506.09985); <code>facebookresearch/vjepa2</code>	Internet-scale video JEPA (~1M+ h) + post-train	CLEAR
V-JEPA 2-AC	Action-conditioned world model on <62 h DROID; zero-shot Franka pick-place via CEM + image goals (no env-specific task reward)	CLEAR robotics transfer
V-JEPA 2.1 (dense predictive loss / multi-modal tokenizers)	Meta repo docs	TENTATIVE -as-of-pack

Design rhyme vs VLA towers (TENTATIVE — do not overclaim)

π_0 / $\pi_{0.5}$ / OpenVLA / GR00T packs under `/workspace/` **do not cite** I-JEPA/V-JEPA. Their “predictive” pieces are **action experts** (flow matching / diffusion) on **SigLIP / DINOv2** tokens — architecture **rhyme** with “predict latent futures,” **not** JEPA lineage. BEHAVIOR’25 winners ride $\pi_{0.5}$ + **FM** + **SigLIP** — a **parallel** world-interface bet, not V-JEPA 2-AC.

World-model exam bite: “I-JEPA = static **within-image** latent multi-block prediction (emb, not pixels); robotics payoff is **CLEAR** only at **V-JEPA 2-AC** (action-cond. + CEM); this PDF has **zero** robot numbers — cite lineage, not Assran’23 SR.”

5. Deep dive — SSL taxonomy (JEPA vs MAE vs JEA)

Callout: Fig. 2 EBM framing is the session's conceptual spine. Three families:

Family	Predict / match	Space	Hand-crafted views?	Off-the-shelf semantic?
Joint-Embedding (JEA) — SimCLR, BYOL, DINO, VICReg, Barlow, iBOT	Similar embeddings for augmented views	Embedding	Yes	High; aug bias; multi-view cost
Generative / MIM — MAE, BEiT, SimMIM	Reconstruct missing patches	Pixels / tokens	No (masking)	Lower linear; needs FT
JEPA (this paper)	Predict target embeddings from context	Embedding	No (mask prior)	High linear + good local; single view

Architecture checklist (Empiricist pin)

Piece	Spec (paper default)
Context encoder (f_{θ})	ViT; only visible context patches; no [CLS] ; eval = avg-pool target-encoder
Target encoder ($f_{\bar{\theta}}$)	Same ViT; full image ; EMA 0.996 → 1.0
Predictor (g_{ϕ})	Narrow ViT emb 384 ; depth 6 (B) / 12 (L/H) / 16 (G)
Targets	Mask output of target-encoder; M=4 blocks scale (0.15, 0.2), aspect (0.75, 1.5)
Context	1 block scale (0.85, 1.0); remove target overlap → sparse informative (~ 0.25 patches)
Loss	$\frac{1}{M} \sum_i \sum_j \in B_i \hat{s}_{y_j} - s_{y_j} ^2$
Opt	AdamW; BS 2048 ; LR $1e-4 \rightarrow 1e-3$ (15-ep warmup) → $1e-6$ cosine; WD 0.04 → 0.4

Ablation Bot AUTHORITATIVE — must-hit deltas

Ablation	Paper locus	Headline numbers
Mask strategy	T6 · ViT-B/16 300ep · IN-1% lin	multi-block 54.2 · raster 15.5 · block 20.2 · random 17.6
Target space	T7 · ViT-L/16	emb 66.9 (500ep) · pixels 40.7 (800ep)
Target scale	T8	sweet spot (0.15,0.2) → 54.2 ; too small 19.2 ; too large 33.6
Context scale	T9	(0.40) 31.2 → (0.85) 54.2
# targets	T10	1 → 4: 9.0 → 54.2
Mask where	T11 · ViT-H/16	output 67.3 · input 56.1
Predictor depth	T12 · ViT-L/16 500ep	6 → 64.0 ; 12 → 66.9
Predictor width	T14 · 1% FT	384 → 70.7 ; 1024 → 68.4 (bottleneck helps)
Data/model scale	T5	IN22k lifts semantic transfer; G/16 patch tradeoff on Clevr

Taxonomy exam bite: “Three SSL poles: **JEA** (view-invariance) · **MAE** (pixel MIM) · **I-JEPA** (emb multi-block prediction); multi-block **54.2**»**15–20**; emb«pixels **66.9 vs 40.7**; Clevr Dist **72.4** keeps local structure JEA discards.”

6. Role-ready sections (all five — Archaeologist RICHEST)

6.1 Archaeologist — WEIGHT 1 LONGEST PRIMARY (Kanta Saito)

Seat charge: Deepen **CLEAR / TENTATIVE** lineage from predictive coding → LeCun AMI / JEPA → **I-JEPA** → **V-JEPA** → **V-JEPA 2 / 2-AC**. Map consistency triangle vs DINOv2/SigLIP and MAE. Explicitly fence robot claims to **CLEAR** subsequent papers. Longest talk block (~16–18 slides).

Priors — CLEAR (cited / direct parents)

Prior	Role relative to I-JEPA	Label
LeCun 2022 AMI v0.9 [48]	Names JEPA ; predictive embeddings + EBM; AMI agenda	CLEAR (cited)
LeCun et al. EBM tutorial [49]	Energy assign compatible/incompatible inputs — Fig. 2 framing	CLEAR
Rao & Ballard / Friston predictive coding [59, 31]	Cognitive motivation: predict missing / latent causes	CLEAR (cited)
Siamese SSL: SimCLR, MoCo, BYOL, SimSiam, DINO, SwAV, MSN, iBOT	View-invariance JEA foil; EMA / stop-grad collapse tricks reused	CLEAR
VICReg, Barlow Twins	Non-contrastive redundancy-reduction JEAs	CLEAR
MAE, BEiT, SimMIM, Context Encoder, DAE	Generative / MIM foil ; I-JEPA reuses sparse-encoder trick	CLEAR
data2vec, Context Autoencoders	Closest: predict representations of masked patches; I-JEPA differs by multi-block + efficiency + image-only scope here	CLEAR
CPC / CPC images	Local predictive coding with contrastive patch discrimination — paper distinguishes: I-JEPA is non-contrastive block prediction	CLEAR (related-work contrast)
ViT / DeiT	Backbone	CLEAR

This paper — CLEAR inventum

1. First **image** instantiation of JEPA with **multi-block** masking + EMA target + narrow **positional** predictor.
2. Demonstrates: **no view augs + embedding-space loss** → semantic linear probes **and** competitive local tasks + **major compute win** vs MAE.
3. Predictor RCDM visualizations as evidence the predictor captures **pose / part structure** while discarding unpredictable detail — philosophical JEPA claim made visual.
4. Mask design **replaces** view-aug invariance as the semantic prior.

Subsequent — CLEAR

Work	What it adds	Label
V-JEPA (2404.08471); facebookresearch/jepa	Video JEPA; feature prediction; strong frozen probes	CLEAR (official follow-on; cites I-JEPA)
V-JEPA 2 (2506.09985); facebookresearch/vjepa2	Internet-scale video + planning path	CLEAR
V-JEPA 2-AC	<62 h DROID → zero-shot Franka CEM + image goals	CLEAR robotics
Official facebookresearch/ijepa	Reproducible IN configs / checkpoints	CLEAR

Subsequent — TENTATIVE / design rhyme (do not overclaim)

Work	Relation	Label
DINOv2	Strongest modern discriminative dense features; complementary foil, often paired with SigLIP in VLAs	Design foil , not child
π_0 / $\pi_{0.5}$, OpenVLA, GR00T-N1	Local packs: no I-JEPA/V-JEPA cites; action experts on SigLIP/DINOv2	Architecture rhyme only
Point-cloud / other-modality JEPA (Meta blog)	Pending pack-local PDFs	TENTATIVE
BEHAVIOR winners ($\pi_{0.5}$ stacks)	FM + SigLIP — parallel world-model bet, not V-JEPA 2-AC	TENTATIVE as “same agenda”

Genealogy diagram (CLEAR framing)

```
Predictive coding (Rao & Ballard '99; Friston)
  → Energy-Based Models / LeCun tutorial [49]
    → LeCun AMI v0.9 (2022) – JEPA abstract [48]
      → data2vec / CAE / MAE (repr vs pixel / token targets; MIM foil)
      → BYOL / DINO / iBOT (EMA / Siamese collapse tricks; JEA foil)
        → **I-JEPA (this paper, CVPR 2023)** – image JEPA + multi-block mask
          → **V-JEPA** (2024) – video feature prediction
            → **V-JEPA 2 / 2-AC** (2025) – action-cond. world model + robot CEM
              → optional visual-backbone *candidate* vs SigLIP/DINOv2 in VLA stacks
                (Empiricist E8; not default ship)
```

Consistency triangle (session tattoo)

```
DINOv2 / SigLIP — discriminative JEA / contrastive-sigmoid — VLA vision towers ( $\pi_{\theta}$ ...)
MAE — generative MIM (pixels) — strong FT, weaker linear
I-JEPA → V-JEPA → V-JEPA 2-AC — predictive embedding / world model — planning robotics
```

Deepen-toward-robotics note (Kanta)

I-JEPA's durable robotics-facing idea is **not** “better ImageNet backbone for BC,” but **predictor-as-world-model**: learn $P(\text{latent}\{target\} | \text{mid}\text{latent}\{context\}, z)$ without wasting capacity on pixel noise. **V-JEPA 2-AC** is the first **CLEAR** Meta demonstration that this scales to **action-conditioned** latent dynamics + zero-shot manip planning. For BEHAVIOR / VLA groups: ask whether an **action-conditioned JEPA head** on frozen video features beats / complements FM action experts — **open research**, not established in this 2023 paper.

What to cite I-JEPA for (2026 talks)

- Semantic SSL **without** SimCLR-style augs.
- **Latent** prediction vs MAE pixels.
- Multi-block masking as the critical inductive bias.
- Efficient scaling narrative (cite Fig 1/5; don't re-run).
- Birth certificate of the JEPA → V-JEPA 2-AC line.
- **Not for**: robot SR, BEHAVIOR q-score, VLA action heads, “ π_{θ} uses JEPA.”

Archaeologist exam bite: “I-JEPA (Assran'23 CVPR) = image JEPA: emb multi-block prediction + EMA + no view augs; multi-block **54.2** $\gg$ **15–20**; emb **66.9** $\gg$ pixels **40.7**; CLEAR line → **V-JEPA** → **2-AC** for robots — **do not** claim robot numbers from this PDF; triangle = DINOv2/SigLIP | MAE | I-JEPA → V-JEPA → 2-AC.”

6.2 Stakeholder — WEIGHT 2 (Lukas Moroz)

Buy / why schedule this paper in “Accompanying Models”:

- Foundational **non-generative** SSL brick that Meta later scaled into **video world models + robot planning** (V-JEPA 2-AC).
- Practical win even for static vision: **ViT-H/14 <72 h on 16 A100**, linear **79.3%**, 1% **73.3%**, without view-aug engineering.
- Dual competence: **semantic** (closes gap to DINO/iBOT) and **local** (Clevr depth **72.4** $\gg$ DINO **53.4**) — less “invariance over-bias.”
- Open code (facebookresearch/ijepa); clear recipe to reimplement / ablate.
- Conceptual clarity: three-way taxonomy **contrastive / generative / predictive** — reduces confused “SSL = contrastive” talk.

Risks / costs / limits:

- Paper itself is **image-only**; no planning / action / robot numbers — value is **lineage + idea**, not a drop-in world

model.

- Linear-probe SOTA for view-aug methods still contested by iBOT/DINO at smaller compute narratives; need scale (H/16@448) to match.
- Full FT still **0.7 pp** behind MAE H/14@448 (87.1 vs 87.8) despite **5.3×** fewer epochs — if KPI is end-to-end FT only, MAE remains fine.
- Predictor + EMA target $\approx$ 7% slower/iter than MAE; savings come from **epoch count**.
- Multi-block hyperparams are **sensitive** (T6–T10): wrong mask $\rightarrow$ collapse to mid-teens 1% accuracy.
- Downstream VLA stacks (π_0 , OpenVLA, GR00T) **did not adopt** I-JEPA encoders — SigLIP/DINO won the “vision tower” market. JEPA’s robotics payoff today is **world-model / planning** (V-JEPA 2-AC), not replacing SigLIP in PaliGemma.

Decision ask: Treat I-JEPA as **required reading for world-model track**; optional for VLA backbone bake-offs unless evaluating latent predictors.

Stakeholder exam bite: “Buy I-JEPA as the **JEPA birth certificate** + efficient semantic SSL recipe; buy V-JEPA 2-AC for robot planning demos — don’t buy I-JEPA as a drop-in π_0 tower replacement.”

6.3 Scientific Reviewer — produced despite empty seat

Claims under review

1. Predicting in **representation space** with multi-block masking yields semantic SSL **without** hand-crafted view augs.
2. Multi-block masking is **critical** ($\gg$ random / single-block / raster).
3. I-JEPA is **more compute-efficient** than MAE at matched semantic quality (GPU-hour plots).
4. Representations remain strong on **local** tasks (Clevr) vs view-invariance methods.

Strengths

1. Clean taxonomy (Fig. 2) and crisp instantiation (Fig. 3).
2. Ablations that isolate the thesis: emb $\gg$ pixels (**66.9 vs 40.7**); multi-block $\gg$ alts (**54.2 vs ≤ 20.2**); output-masking $\gg$ input-masking.
3. Reports both **semantic** and **local** probes — rare honesty vs invariance-biased SSL.
4. Efficiency claims backed by GPU-hour plots (Figs. 1, 5), not only epochs.
5. Ablation Bot corpus rich (T6–T14 + App C).

Weaknesses / ask-for-revision style

1. “No hand-crafted augs” still uses **block sampling** as inductive bias — soft-aug debate.
2. Table 2 mixes FT vs linear for 1% (“whichever works best”) — hurts strict ranking.
3. Predictor viz via RCDM is qualitative only.
4. Limited failure-mode analysis (texture bias, long-tail, multi-object beyond Clevr).
5. Closest concurrent (data2vec) comparison could be tighter on matched compute.
6. IN-centric ablations — “semantic” $\approx$ ImageNet class features; robot OOD untested here.
7. Compute claims mix architectures (H/14 vs S/16 iBOT) — directionally persuasive, not iso-param.
8. Collapse theory thin: EMA “essential” empirically; limited analysis without it.

Verdict: Strong **methods** paper that cleanly carves the third SSL pole and seeds the JEPA program. Accept for Accompanying Models as **foundational reading**; when citing for robotics, demand **V-JEPA 2-AC**, not this paper’s experiments.

Reviewer exam bite: Demand **T6 + T7 + Figs 1/5** before slogan; caveat image-only scope; punish “I-JEPA did robot planning” and “course smoke = 79% H/14.”

6.4 Empiricist — produced despite empty seat (course smoke; all seats consume)

Seat charge: Course smokes that stress **mask strategy · emb vs pixels · linear probe**, ranked by GPU honesty. Prefer CIFAR-100 / IN-100 / tiny IN subset + ViT-Tiny/S — not paper ViT-H/14 IN-1K 300ep @ BS2048.

Claim under measurement

Matched short pretrain: **multi-block** masking and **EMA embedding** targets beat random/block masks and pixel reconstruction on semantic linear / k-NN probes (directional T6/T7), without claiming paper absolute %.

Hypotheses (falsifiable — from DELIVERABLE)

1. **H-mask**: On tiny data, multi-block still beats random / single-block on linear/k-NN after matched short pretrain (qualitative T6).
2. **H-space**: Matched arch+steps: predicting **EMA target embeddings** beats **pixel reconstruction** on semantic probes; pixel path may win a low-level toy — falsifiable split (T7 vs T4 tension).
3. **H-predictor**: Too-shallow predictors underfit positional conditioning → worse probe; depth helps until compute wall (T12 direction).
4. **H-transfer**: After smoke pretrain, linear probe on CIFAR100 / IN-100 rises above train-from-scratch supervised tiny baseline at same steps; absolute % **will not** match paper H/14.
5. **H-ckpt (E8)**: Public I-JEPA ViT features remain competitive with MAE on household robot stills for linear/retrieval probes — optional bridge to BEHAVIOR perception.

Falsifiers: multi-block $\not>$ random; emb $\not>$ pixels on semantic probe; depth flat/negative; probe $\leq$ scratch; public JEPA $\ll$ MAE on robot stills with no caveat.

Prioritized ablation matrix (Ablation Bot + DELIVERABLE E1–E11)

#	Experiment	Paper hook	Smoke	Pri
E1	Mask: multi vs random vs block (vs raster)	T6	Y — PRIMARY	P0
E2	JEPA emb+EMA vs pixel decoder	T7	Y — PRIMARY	P0
E3	Target / context / #targets mini-grid	T8–T10	Y	P0
E4	Predictor depth {2,4,6}	T12	Y	P1
E5	Mask output vs input	T11	Partial	P1
E6	Frozen linear / k-NN first	T1–T3 lite	Y — PRIMARY	P0
E7	Clevr or toy count/depth	T4	Partial	P2
E8	Frozen public ckpt on robot/household stills	Transfer hook	Y	P1
E9	EMA off / stop-grad	§3	Y	P2
E10	Full ViT-H/14 IN-1K 300ep BS2048	Fig1/5	N	Skip
E11	IN-22K / ViT-G	T5	N	Skip

Min package: E1 + E2 + E6, then E3 if GPU free; E8 as Archaeologist/Empiricist joint measurement with public weights.

Exact reproduce stack (EDIT)

```
# Official
git clone https://github.com/facebookresearch/ijepa
# Pin commit → logs/pins.txt

# Course scout env (quiet outlines – preferred)
export IJEPa_WORK_ROOT=$SCRATCH/ijepa-smoke
cd /workspace/ijepa-quiet
bash env_outline/setup_mamba.sh
# mamba activate ijepa-smoke
mkdir -p logs runs checkpoints configs
# EDIT partition/account in slurm_*.sh
# Outlines only – do NOT sbatch from quiet box
# NEVER claim paper 79.3% H/14 / <1200 GPU-h / 10× MAE from smoke
```


Run cards (from DELIVERABLE)

E1 — `mask_strategy` (PRIMARY)

Field	Spec
Paper	multi-block 54.2 vs raster 15.5 / block 20.2 / random 17.6
Goal	multi-block > random & block on tiny linear/k-NN by clear margin
Train	ViT-S/16 or Tiny; CIFAR100@224 or IN-100; BS 256–512; ~20k–80k steps
Pass	3 mask runs logged; directional T6; not 54.2 absolute
Budget	~6–18 GPU-h · VRAM 8–16 GB
Script	<code>env_outline/slurm_e1_mask.sh</code>

E2 — `jepa_vs_mae_proxy` (PRIMARY)

Field	Spec
Paper	emb 66.9 vs pixels 40.7
Goal	emb+EMA > pixel decoder on semantic probe
Pass	Matched mask geometry + steps; probe delta in JEPA direction
Budget	~6–18 GPU-h
Script	<code>env_outline/slurm_e2_jepa_vs_mae.sh</code>

E6 — `linear_probe` (PRIMARY)

Field	Spec
Paper	T1–T3 protocol
Goal	Frozen probe / k-NN every N steps; beats scratch tiny baseline
Pass	Probe curve logged; disclaimer vs paper H/14
Budget	0.5–3 GPU-h (frozen)
Script	<code>env_outline/slurm_e6_linear_probe.sh</code>

E3 / E4 / E8 — follow-ons

E3 mini-grid (4–6 short runs) after E1 scaffold; E4 predictor depth {2,4,6}; E8 frozen public I-JEPA vs MAE/DINO/SigLIP on BEHAVIOR/DROID stills (linear/retrieval) — ties Accompanying Models → VLA stacks **without** pretrain.

Numbers to memorize (exam)

Fact	Value
Multi-block / raster / block / random	54.2 / 15.5 / 20.2 / 17.6
Emb / pixels	66.9 / 40.7
IN linear H/14 / H/16@448	79.3 / 81.1
IN-1% H/14 / H/16@448	73.3 / 77.3
CIFAR100 / Places / iNat (H/14)	87.5 / 58.4 / 47.6
Clevr Count / Dist	86.7 / 72.4
Output / input mask	67.3 / 56.1
Pred depth 12 / 6	66.9 / 64.0
FT I-JEPA / MAE	87.1 / 87.8 (300 vs 1600 ep)
Compute	<1200 GPU-h; 16×A100 <72 h; ~10× vs MAE

Failure modes cheat-sheet

Symptom	Likely cause	Next action
All masks ~same	Steps too few / data too tiny	Extend steps; try IN-100
Emb >pixels	Pixel path undertrained or wrong eval	Match steps; semantic probe only
Collapse (probe ~chance)	EMA off / LR too high	Restore EMA; check mom schedule
OOM	BS 2048 paper	BS 256–512; ViT-S/Tiny
Claimed 79% from smoke	Overclaim	Report directional only
E8 JEPA << SigLIP on language retrieval	Expected	JEPA unlabeled; pair projector or keep SigLIP default

What not to invent / claim

- Do not claim course smoke = paper H/14 79.3% / <1200 GPU-h / 10× MAE.
- Do not claim robot SR / BEHAVIOR wins from this paper.
- Do not launch E10/E11 from course queue.
- Do not compare smoke CIFAR % to paper IN linear without disclaimer.
- Do not claim π_0 /OpenVLA use I-JEPA towers.

Empiricist exam bite: “Assran’23: multi-block 54.2>>15–20; emb 66.9>> pixels 40.7; Empiricist E1+E2+E6 on tiny data for direction — never claim H/14 IN parity or robot SR from smoke / this PDF.”

6.5 Visionary — produced despite empty seat (BEHAVIOR recipe; no Discord/Drive)

BEHAVIOR connection:

2025 BEHAVIOR winners ride $\pi_{0.5}$ + flow-matching action experts + SigLIP/PaliGemma — a generative action world-interface, not a JEPA latent dynamics model. I-JEPA → V-JEPA 2-AC is the parallel bet: plan by optimizing actions so predicted latents match a goal image embedding, with almost no task demos. Opportunity: hybrid — V-JEPA-style latent rollout for long-horizon feasibility / imagination; FM expert for short-horizon motor execution.

BEHAVIOR need	I-JEPA-shaped response
Visual trunk choice	Default stays SigLIP-class; I-JEPA = ablation candidate (Empiricist E8)
Semantic ID w/o language	Try frozen JEPA features where view-aug SSL biases hurt
Long-horizon imagination	Invest in video JEPA / V-JEPA 2-AC, not static I-JEPA alone
Goal-image planning	CEM on action-cond. latents (cite 2-AC, not this PDF)
Eval honesty	Encoder bake-offs on challenge frames — citing 79% IN does not move q-score

CLEAR: Design prior that latent multi-block prediction is a viable SSL path and seeds world-model research.
TENTATIVE: Official BEHAVIOR’26 baseline ships JEPA trunk; hybrid JEPA-rollout + FM wins organizer board.

2026 BEHAVIOR visual-backbone / world-model recipe (6 bets)

1. Default challenge trunk stays SigLIP-class (OpenPI) or organizer N1.7 vision — I-JEPA is an ablation candidate, not the default ship.
2. When to try JEPA features: tasks heavy on semantic object identity without language, or domains where view-aug SSL biases hurt (unusual wrist fisheye / sim materials) — validate with Empiricist E8 before any FT.
3. Prefer frozen-probe → LoRA encoder FT over from-scratch I-JEPA pretrain on BEHAVIOR RGB (E10-scale compute is politics, not student path).
4. Do not replace language alignment blindly: BEHAVIOR instructions / stage labels still favor SigLIP / VLM towers; JEPA is unlabeled-image SSL — pair with a language projector if used.
5. World-model path ≠ this PDF: for stage-aware long horizon, invest in video JEPA / latent dynamics siblings + RLC-style stage structure — not static I-JEPA alone.
6. Honesty: citing Assran et al. 79% IN linear does not move BEHAVIOR q-score; only encoder ablations on challenge data do.

Follow-up research

- 1. Action-conditioned I/V-JEPA on OXE / DROID / BEHAVIOR-1K video — sample efficiency vs π_0 -style BC.
- 2. Goal-image CEM planning (V-JEPA 2-AC recipe) inside BEHAVIOR scenes vs language-goal VLAs.
- 3. Uncertainty-aware predictors for contact-rich failure anticipation.
- 4. Cross-embodied latent predictor (camera + proprio tokens).
- 5. Contrastive-vs-JEPA bake-off for **affordance** probes.

New applications (beyond Discord/Drive)

- Zero-shot rearrange with **image goals** on new kitchens (V-JEPA 2-AC recipe).
- Video anticipation / anomaly for safety monitors (predictor residual as surprise).
- Self-supervised pretrain for **simulation-free** world models in household robotics.
- Educational: teach “representations should be predictive, not reconstructive” as the AMI slogan made concrete.

Visionary one-liner: I-JEPA is the **static-image birth certificate** of Meta’s world-model line; the actionable robotics paper is **V-JEPA 2 / 2-AC** — but you cannot understand that line without this one. For BEHAVIOR’26, treat I-JEPA as a **visual-backbone ablation** against SigLIP/DINO/MAE while policy glue stays chunked FM / $\pi_{0.5}$ / N1.7.

Visionary exam bite: “For BEHAVIOR, default SigLIP; ablate **I-JEPA** features via E8; world-model ambition rides **V-JEPA 2-AC**, not this still-image PDF — **no Discord/Drive.**”

(No Discord / Drive checklist content.)

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export IJEPA_WORK_ROOT=$SCRATCH/ijepa-smoke
cd /workspace/ijepa-quiet
bash env_outline/setup_mamba.sh
# mamba activate ijepa-smoke
mkdir -p logs runs checkpoints configs
# EDIT: partition/account in slurm_*.sh
# Pin facebookresearch/ijepa commit → logs/pins.txt
# Outlines only – do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	ijepa-smoke sketch
env_outline/conda_env.md	Human-readable mamba steps
env_outline/setup_mamba.sh	Create env + clone pointer
env_outline/run_checklist.md	GPU honesty preflight
env_outline/slurm_e1_mask.sh	Mask-strategy array — E1 PRIMARY
env_outline/slurm_e2_jepa_vs_mae.sh	Emb vs pixel — E2 PRIMARY
env_outline/slurm_e6_linear_probe.sh	Frozen probe — E6 PRIMARY
env_outline/README.md	Index + honesty rules; siblings

GPU / RAM / TIME budget honesty

Assumptions: ViT-Tiny/S/16, CIFAR100 or IN-100, BS **256–512** (paper **2048**), short steps — **not** ViT-H/14 IN-1K 300ep / 16×A100.

Workload	GPU VRAM (peak)	Host RAM	Wall time (order)	Smoke
E1/E2 ViT-S, CIFAR100, ~50k steps	8–16 GB	32–64 GB	6–18 h	Y
E3 mini-grid (4–6 short)	same	same	+0.5–1× E1	Y
E4 predictor depth ×3	same–slightly less	same	+few h	Y
E5 output vs input mask	12–24 GB (40GB pref.)	64 GB	overnight	Partial
E6 linear probe only	4–10 GB	32 GB	0.5–3 h	Y
E8 frozen public ckpt on robot stills	6–12 GB	32 GB	1–4 h	Y
E7 Clevr	8–16 GB	64 GB	data-bound	Partial
Paper ViT-H/14 IN-1K 300ep BS2048	16×A100	node-class	<72 h wall / <1200 GPU-h	N
IN-22K / ViT-G	multi-node	—	days+	N

Honesty rule: `log (arch,mask,target_space,pred_depth,steps,BS,probe,wall_s,mem)` . If `#SBATCH --gres=gpu:N` but `CUDA false` → **fail**. Never amortize paper **79.3% / 10× MAE** onto a Partial stub.

Course envelope: target ≤ ~40–60 GPU-h for E1–E4+E6; reserve E8 if public weights cached. Paper H/14 story = cite only.

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
Multi-block critical (T6)	E1: multi > random & block on tiny probe	Matching 54.2%
Emb ≫ pixels (T7)	E2: emb+EMA > pixel on semantic probe	Matching 66.9 vs 40.7
Context/targets scales (T8–10)	E3: default-ish beats extremes	Full App C grid
Deeper predictor helps (T12)	E4: depth↑ improves or plateaus	Need depth-12 on ViT-L
Strong linear w/o view augs (T1)	Smoke probe beats scratch; E8 vs DINO frozen	79.3% H/14
Faster than MAE / iBOT (§7)	Optional steps-to-threshold	10× GPU-h claim
Local tasks competitive (T4)	E7 optional toy	Full Clevr parity
Scales to H/14 in <72 h on 16×A100	Cite only	Re-running it

Pass: Empiricist executes **E1 + E2 + E6** with paper-aligned **directional** rankings + GPU honesty logged.

Fail / overclaim: “reproduced Assran’23 / matched 79.3% / verified 10× MAE / robot planning from I-JEPA” from smoke or this PDF alone.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Archaeologist	1 · LONGEST · ~16–18	CLEAR/TENTATIVE lineage; genealogy; consistency triangle; V-JEPA → 2-AC fence; inventum; exam bite
Stakeholder	2 · ~12–14	Buy world-model track; risks (image-only; VLAs didn’t adopt); decision ask
Reviewer	empty seat · ~12–14	T6+T7+Figs1/5; verdict foundational Accept; robotics cite 2-AC
Empiricist	empty seat · ~12–14	H-mask/H-space; E1–E6 run cards; GPU honesty; exam numbers
Visionary	empty seat · ~12–14	BEHAVIOR SigLIP default + JEPA ablation; V-JEPA 2-AC hybrid; no Discord/Drive

Spine (paste into Slide Maker) — 5 bullets:

- Predict in embedding space, not pixels** — multi-block context → target JEPA; no SimCLR view augs; EMA + asymmetric predictor.
- Must-cite:** multi-block **54.2** vs ~15–20; emb vs pixels **66.9** vs **40.7**; IN1K H/14 **79.3** / H/16@448 **81.1**; 1% **73.3/77.3**; Clevr Dist **72.4**; compute **<1200 GPU-h / 16×A100 <72 h / ~10×** vs MAE.
- SSL triangle:** DINOv2/SigLIP (JEA towers) | MAE (pixel MIM) | **I-JEPA → V-JEPA → 2-AC** (predictive emb / world model).
- Lineage (Archaeologist):** LeCun AMI → I-JEPA inventum → **CLEAR** V-JEPA → **V-JEPA 2-AC** robotics — **do NOT**

claim robot results from this PDF.

5. **BEHAVIOR’26 + honesty**: default SigLIP; I-JEPA = backbone ablation (E8); Empiricist E1+E2+E6 directional only; world-model ambition = video JEPA / 2-AC — **no Discord/Drive**.

GitHub title sketch: [ECE 605] - I-JEPA Archaeologist Making One

Full Slide Maker brief: /workspace/ijepa-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/ijepa-quiet/KANTA_PACK.md
PDF	/workspace/ijepa-quiet/KANTA_PACK.pdf
Slide brief	/workspace/ijepa-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/ijepa-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/ijepa-2301.08243-summary.md
Ablations	/workspace/papers/ijepa-2301.08243-ablations.md (AUTHORITATIVE)
Paper notes	/workspace/ijepa-quiet/paper_notes.md
Env outlines	/workspace/ijepa-quiet/env_outline/
Paper PDF	/workspace/papers/ijepa-2301.08243.pdf
Page figs	/workspace/papers/ijepa-figs/page-*.png

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 — Nov 25 Accompanying Models · Archaeologist=1 (Kanta) LONGEST · Stakeholder=2 · all five produced · Ablation Bot AUTHORITATIVE · no Discord/Drive.